



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: COURSE: DATA ANALYTICS AND VISUALIZATION - 24DSB3101 A.Y. 2026 - 2027

TITLE OF THE PROJECT	Intelligent customer churn prediction using data analytics and classification models	
S. No.	University ID	Name
1	2420030540	Allipuram Akhil
2	2420030440	D.Surya
Name of the Guide		Dr.K.Swapnika

Abstract

Customer churn represents one of the most critical challenges confronting modern service-based organizations, since the cost of retaining an existing customer is substantially lower than the cost of acquiring a new one. This project, titled *Intelligent Customer Churn Prediction Using Data Analytics and Classification Models*, aims to develop a data-driven system capable of identifying customers who are at risk of discontinuing their relationship with an organization. Publicly available customer datasets containing demographic, account, service-usage, contract, and billing information form the foundation of the study. The proposed methodology begins with rigorous data preprocessing, including missing-value treatment, categorical encoding, feature scaling, and class-imbalance handling, followed by a comprehensive **Exploratory Data Analysis (EDA)** to uncover behavioral patterns and the relationships between customer attributes and churn outcomes. Domain-informed **Feature Engineering** is then applied to derive meaningful indicators such as tenure groups, charge ratios, and service-usage combinations that strengthen the predictive signal available to downstream models. Multiple **Classification Models**, including Logistic Regression, Decision Tree, Random Forest, XGBoost, and CatBoost, are trained and rigorously compared using Accuracy, Precision, Recall, F1-Score, and ROC-AUC, ensuring that performance is assessed fairly despite the inherent imbalance of churn datasets. The resulting insights are translated into interactive visualizations and a dashboard that highlight high-risk customers and the most influential churn factors, enabling stakeholders to interpret model outcomes without requiring technical expertise. By combining **Predictive Analytics** with transparent, actionable visual reporting, this project seeks to support organizations in designing targeted, data-driven customer retention strategies that reduce attrition and preserve long-term customer value.

Signature of the Guide

HOD

